

: Exqutor : AnalyticDB-V; A Hybrid Analytical Engine Towards Query Fusion for Structured and Unstructured Data (VLDB 2020, Alibaba Group) : 2026 3 : ,

가 , , 가
MySQL PostgreSQL 가
RDBMS FAISS Milvus 가
“ , 10 가 ” ?
 , 가
AnalyticDB-V Alibaba Group
 . 가

AnalyticDB-V . , SQL 가 0.5 가 (predicates)가 WHERE price < 100000 AND category = 'electronics' AND stock > 0 가 (selectivity) . “ 0.5 가 , . 가 VAQ(Vector-Augmented Analytical Query) . AnalyticDB-V SQL .

```
SELECT product_id, price, similarity_score
FROM products
WHERE category = 'electronics'
AND price < 100000
AND ANNS(image_embedding, query_vector, top_k=100, accuracy=0.95)
ORDER BY similarity_score DESC;
```

AnalyticDB-V Lambda .

(Streaming Layer) 각 . ,가 . . 각 . , .

(Batch Layer) (: 3) , . . , VG PQ ANN . ,가 가 나 . . , , , .

Lambda . , (freshness) .가 . , . , .

Lambda , AnalyticDB-V VG PQ(Vector Graph Product Quantization) . IVF-PQ .

IVF-PQ .가 k-means () . 1 1 , 1 가 .가 , 10 10 100 .

IVF-PQ . , () , k-means () . , 3 , 50 .

VG PQ . (anchor points) . , - , . ,가 , - .

VG PQ IVF-PQ 가 , ,가 . , VG PQ IVF-PQ 2-3 .

1.4 정확도 기반 비용 최적화 (Accuracy-Aware Cost-Based Optimization)

AnalyticDB-V . . (execution plan) 가 . I/O , CPU , .

1: price < 100000 (1000) , : 1000 + .

2: 100 , price < 100000 . : 100 + 100 . 2 .

(approximate) . ,가 . , VG PQ 가 .

AnalyticDB-V . .가 가 .

- (: 95%)
-
-

가 .

가 .가 “95% ” , 99.9% . ,가 99.99% 가 , .

,가 ANNS (accuracy function) .

= f(, ,)

AnalyticDB-V , .

(k) = 1.0 - 0.5 * exp(-k/500)

k ()가 가 . k=100 63%, k=1000 99% .

1.5

AnalyticDB-V 가 .

, WHERE price < 1000 ,

.

• 가 (: 100), + .

• 가 (: 100), .

• 가 (: 5000), .

, “1000 < price <= 2000 1 ” , WHERE price < 2000 .

. q가 , “가 d ” . ?

,가가 .가 , .

,가 .가 d가 , .

, (,) .가 “ ” .

AnalyticDB-V , , 가 , “ (: 10%)”가 .가 ,

.

,가 . :

SELECT *

FROM products

WHERE image_similarity(embedding, query_img) < threshold

AND price < 100000

1: , - : 100 1000 -가 : 1000 300가 - : + 1000

2: , -가 : 100 30 - : 30 100 - : 30 +

?가 .가 1000 1 (1000 > 300) .

가

(selectivity)가 가 , WHERE A AND B selectivity(A) × selectivity(B) .
가 ,가 , “가 ” .

1.6 가

AnalyticDB-V Alibaba .

:- : 5000 (512 CNN)- : , , , , - : VAQ (+ 가 +
+)- : (Faiss), DB(MySQL),

:

:- (top-k) IVF-PQ VGPQ 2-3 - 100 top-100 10ms

:- + : 50 (:)- + + + : 100 - 가 5-20

: - 99.9% : 100ms - 99% : 50ms - 95% : 20ms - 90% : 10ms

1.7 가 Exqutor 가

AnalyticDB-V ,가 .

AnalyticDB-V ,가 (: 10%) , , 10%가 .
 , .

, (: “ ”) ,가 (30%가) . ,
(: “ ”) ,가 (5%) .

,가 가 (30-40%)가 , (1-2%) .

AnalyticDB-V ? 가 .

1. : , .

2. : , .

3. : ,가 , .

Exqutor , AnalyticDB-V .

Exqutor . AnalyticDB-V 가 가 가, Exqutor . , VGPQ 가

1. :가

2. :

3. - :

“Extended Query Optimizer” . AnalyticDB-V (extended) ,가

.

2. 쿼리 쿼리 쿼리 (쿼리)

2.1 VAQ(Vector-Augmented Analytical Query)란?

가 VAQ가 ? ?” ?

VAQ가 SQL .가 .

,가 . “ ,가 15 4.5 ” .

():

. “가 0.3 ”. 10,000 .

, . “ 10,000 ID , < 150000 AND >= 4.5 ”. 300 .

:

1. : 10,000

2. :가 ,

3. :가

4. :

VAQ (, AnalyticDB-V):

SQL :

SELECT product_id, price, rating, similarity_score

FROM products

WHERE price < 150000

AND rating >= 4.5

AND ANNS(shoe_image_embedding, query_embedding, top_k=1000, accuracy=0.95)

ORDER BY similarity_score **DESC**

LIMIT 300;

:

1. : ,

2. :가가 ,

3. :가가

4. : 95% ,가

VAQ - . “ ” , “ ” .

2.2 Lambda 쿼리: 쿼리 쿼리 쿼리

“ Lambda ?” .

Lambda . AnalyticDB-V 가

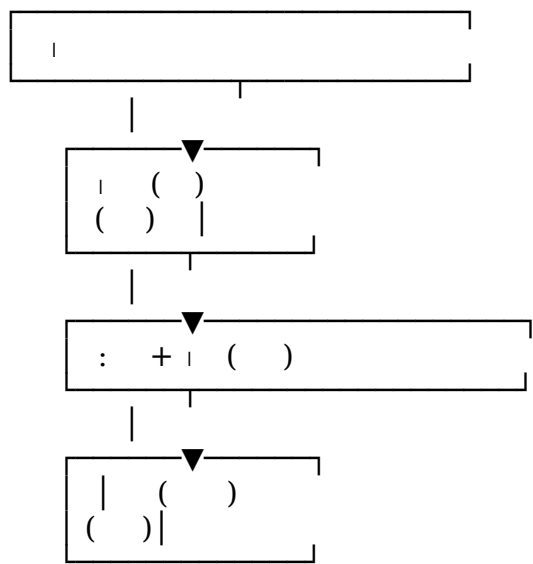
1: (Freshness)

Alibaba 가 , 5 . 가 1 , .

2: (Efficiency)

, ANN . k-means , .

Lambda :



(Streaming Layer):

. 가 (: , B-tree) . , .
. , 1 .

(Batch Layer):

(: 3), . :

- 1.
- 2.
- 3. VG PQ
- 4. ,

(: 가 가). 가 , .

(Query Processing Layer):

, 가 :

- 1. :
- 2. →
- 3. →
- 4. : , ,

5.

Lambda :

1. :
2. :
3. :
4. :가 가 ,

Lambda :

1. : ()
2. :가
3. :

2.3 VGPQ(Vector Graph Product Quantization) □□□□

“VGPQ IVF-PQ ?” .

가 . 1 1000 1 .

IVF-PQ (Inverted File with Product Quantization) - :

IVF-PQ : :

- 1 : 가 (Indexing) - k-means K (: K=10,000) - “ ” - 가
- 2 : Product Quantization - 가 (: 512 4 128) - (quantization) - 가
(: 512 float32 = 2KB)
- :- - 가 (1) - , - 10 (10) - 1000

IVF-PQ :

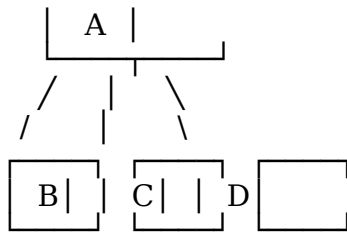
1. :가 , .
2. : 512 , k-means . “ ” ,가 .
3. : K , 3 , 100 . .
4. : .

VGPQ (Vector Graph Product Quantization) - AnalyticDB-V :

VGPQ IVF-PQ .

:

□



“ - ” . , A 100 - , - 1000 가 .

:

- 1.
2. :가 (, : 20)
3. -
4. : -가 →
- 5.
6. 1000

VGPQ :

1. : , “ ” .
2. : , .
3. : - . - , .
4. :가≡ - , .
5. - : - , .

:

- IVF-PQ 2-3
- 512 5000 top-100 : 10-15ms (IVF-PQ 30-40ms)
- 99% : VGPQ 1000 , IVF-PQ 5000

2.4 □□□ □□ □□ □□ □□□

“가 ?” .

. I/O , CPU , .

:

SELECT * FROM products
WHERE ANNS(embedding, query_vec) **AND** price < 100000

1: - ANNS : 10,000 (:가 T1) - 가 : 10,000 3,000 (:가 T2) - : T1 + T2

2:가 -가 : 30 (:가 T2') - ANNS: 30 (:가 T1') - : T2' + T1'

"T1 + T2 < T2' + T1' 1 , 2 "

(ANNS) (approximate) .

, VGPQ :

- 100 : 90%, 10ms
- 1000 : 95%, 50ms
- 10000 : 99%, 200ms

. .

" 95% " , 95% 가 .99% , .

,가 " 99.9% " , .

:

가 :

1. : ANNS ,

accuracy(search_cells) = 1.0 - 0.5 * exp(-search_cells/500)

2. : ()

SELECT * FROM products
WHERE ANNS(embedding, query_vec, accuracy=0.95)

3. :가

```
candidates = [ 1, 2, 3, ...]  
feasible = [p for p in candidates if p.accuracy >= user_accuracy_requirement]  
best = min(feasible, key=lambda p: p.cost)
```

:

: 95%

A: (99%) + 가 → 100ms B: (95%) + 가 → 50ms C: 가 + (99%) → 150ms

.가 B . :50ms 95% .

가 " 99% " , B . A C A → 100ms.

:

1. : ,

2. : (:) 90% , 5

3. :

4. SLA : SLA ,

2.5 AnalyticDB-V

“ ?” .

까 ?

가 가 100 100 ” , .

```
SELECT * FROM products p
JOIN orders o ON p.id = o.product_id
WHERE p.price < 1000
```

A: WHERE p.price < 1000 100 → 100 orders → **Nested Loop Join**
(100)

B: WHERE p.price < 1000 100 → 100 → orders → **Hash Join** (1)

B 가 A , 100 .

:

WHERE price < 1000

price :- : 100, 가: 1,000,000 - : 0-100: 1000 , 100-500: 5000 , 500-1000: 3000 , ...

price < 1000 : 9,000 .

, :

SELECT COUNT(*) FROM products WHERE price < 1000

:

WHERE ANNS(embedding, query_embedding, distance < 0.5)

까 ?

:

1. :
 - 1 2 (: 가 0-1M ,)
 - 512 .
2. :
 - 가 , 0.5 1 가
 - , 0.5 10
 - 100 !

3. :
 - 가 ()
 - , 512 22-24 ()
 - 까지

4. :
 - VGPQ IVF-PQ . 0.5
 - “ 0.5 가 , “ 0.5 ”
 -

5. - :
 - ,
 -

AnalyticDB-V :

AnalyticDB-V . (selectivity)가 (: 10%)가 .

$$() = \times$$

$$: 1,000,000 \times 10\% = 100,000$$

:

1. :
2. : 1% , 50%
3. :

:

: (512) - () - ()

A: - (< 0.5): 50,000 (5%)

B: - (< 0.5): 500 (0.05%)

AnalyticDB-V (10%):- A: 100,000 → 2 → - B: 100,000 → 200 →

Exqutor .

3. Exqutor □□ □□□□

3.1 AnalyticDB-V 가 VAQ □□□ □□ □□ □□

AnalyticDB-V Exqutor 가 . , AnalyticDB-V .

, VAQ :

Alibaba . , .

, Lambda :

가 .

, ANNS SQL :

SQL .가 .

, - :

ANN . , .

3.2 “ ” Exqutor

AnalyticDB-V . AnalyticDB-V , .

:

, AnalyticDB-V 가 :

```
SELECT *  
FROM products p  
JOIN user_history h ON p.id = h.product_id  
WHERE ANNS(p.embedding, query_embedding, distance < threshold)  
AND h.date > '2026-01-01'
```

:

1: ANNS - ANNS : 1,000,000 top-k : ANNS - :

2: - : 1,000,000 100,000 (10%) - ANNS: 100,000 -

1 2 ANNS .

AnalyticDB-V : “ANNS 10%” - 1: 100,000 x = C1 - 2: 100,000 + ANNS = C2

.

:

1 ():- ANNS : 50,000 (5%) - 1 ()

2 ():- ANNS : 500 (0.05%) - 2 ()

가 , .

Exqutor :

Exqutor :

(ANNS)

↓

ECQO (Extended Cardinality Query Optimizer)

↓
↓
가
↓
↓

- :
1. : VGPQ ,
 2. : 가
 3. - : , .

3.3 :

AnalyticDB-V 가 .

:
:- , 1000 - 가 (512 ResNet) - 가 (, , , ,)
: , :

```
SELECT product_id, price, rating, stock
FROM products
WHERE ANNS(image_embedding, uploaded_image) AND
category IN ('bags', 'accessories') AND
price BETWEEN 50000 AND 500000 AND
rating >= 4.0 AND
stock > 0
ORDER BY similarity_score DESC
LIMIT 100;
```

AnalyticDB-V :

(): - ANNS: 10% → 100 - category : 5% () → 100 5 - price : 40%
() → 400 () - rating : 60% () → 600 - stock : 90% () → 900

:- : 100 × 10% × 5% × 40% × 60% × 90% = 10,800 ()

“ 1 가 ” , :- (가 가 가) - ANNS

(Exqutor):

:- ‘bags’ ‘accessories’: 45,000 - 50K-500K: 30,000 - >= 4.0: 20,000 -
> 0: 15,000

15,000 ANNS :- , “ ” 500 (3.3%) - : 500

:

- :
1. AnalyticDB-V 7, .
 2. , , .
 3. (), ().
 4. 가 .
- Exqutor : 1. 가 . 2. . 3. 가 .

4. Q&A

Q1. “AnalyticDB-V 가 FAISS 와 비교 어떤가요?”

A: AnalyticDB-V Alibaba Group VLDB 2020 . SQL .
(FAISS) , 가 7 SQL . AnalyticDB-V .
“ , 가 10 ” SQL :

```
SELECT * FROM products
WHERE ANNS(image_embedding, query_image)
AND price < 100000
```

Q2. “VAQ 가 어떤가요?”

A: VAQ Vector-Augmented Analytical Query . “가즈 ” .
, SQL :

```
-- SQL
SELECT * FROM products WHERE price < 100000
```

```
-- VAQ (가즈)
SELECT * FROM products
WHERE price < 100000
AND ANNS(embedding, query_vector) --
```

VAQ 가 , 가 - (,) (,) , .

Q3. “**가격이 1000 이하인 상품들 중에서 가장 저렴한 상품을 찾아주세요**?”

A: **SELECT** product_id, price
FROM products
WHERE price < 1000
ORDER BY price
LIMIT 1;

WHERE price < 1000

() , 3000
:

WHERE ANNS(embedding, query_embedding, distance < 0.5)

?

1. : 512 개 L .
2. : 가 1/2 개 (0.5) , 개 100 . 1 , 100 .
3. : 가 , 개 가 .
4. : VGPQ 가 1/2 , .

AnalyticDB-V (: 10%) .

Q4. “**VGPQ 가 IVF-PQ 와 비교하여 어떤 장점이 있는지를 설명해주세요**?”

A: . IVF-PQ , VGPQ AnalyticDB-V .

IVF-PQ : 1. k-means 가 2. 3. candidate

IVF-PQ :- - (512) k-means - 3 , 100 ()

VGPQ : 1. , 2. - 3. 가 4. 가

:- 2-3 - 512 5000 : top-100 10-15ms (IVF-PQ 30-40ms)

Q5. “**Lambda 와 관련된 개념을 설명해주세요**?”

A: Lambda .

1: (Freshness) - - 5

2: (Efficiency) - ANNS -

Lambda :

: ()

↓

: ()

↓

: + ()

:- 10 : 100 개 → - 2 : → 1 VGPQ → 6 - →
 (10) + (1)

Q6. “가 100 개 100?”

A: AnalyticDB-V .

(approximate) , :

100 → 90% → 10ms
 1000 → 95% → 50ms
 10000 → 99% → 200ms

:

“ 95% ” , 95% 가 , 50ms (1000) .

“ 99% ” , 200ms .

:- : 5-10 - : - :가

Q7. “Exqutor 100 개 100?”

A: AnalyticDB-V Exqutor .

AnalyticDB-V :- VAQ - ANNS SQL -

AnalyticDB-V :- 가 -

Exqutor : ECQO(Extended Cardinality Query Optimizer)

.

AnalyticDB-V Exqutor . AnalyticDB-V , Exqutor .

Q8. “AnalyticDB-V 100 개 100?”

A: AnalyticDB-V , 개 .

1. (가 가 100):

-
 - 100
 -

2. :

- Alibaba e-commerce
 - (,)

3. :

- +
 - ,

4. :

-

- 가
5. :
- 가
 -

Q9. “pgvector (pgvector)가 뭐예요?”

A: AnalyticDB-V pgvector, Milvus 가 :

pgvector (PostgreSQL): - PostgreSQL 가 - SQL - AnalyticDB-V - ANNS

Milvus (DB): - - VAQ

AnalyticDB-V: - (,) - - Alibaba

Exqutor : - AnalyticDB-V + - Exqutor

Q10. “pgvector (pgvector)가 뭐예요?”

A: AnalyticDB-V .

: - Alibaba : 5000 - (512) - : , , , ,

:

: - VGPQ vs IVF-PQ: 2-3 - 100 top-100 : 10-15ms

VAQ() :- : 100 - + 가 : 50 - + / / : 100

- : - 99.9% : 100ms - 99% : 50ms - 95% : 20ms - 90% : 10ms

: , 5-10 .

Q11. “Exqutor 가 뭐예요?”

A: AnalyticDB-V Exqutor :

AnalyticDB-V ():

가

↓

(: 10%)

↓

↓

가

Exqutor ():

가

↓

ECQO 가

↓

↓

↓

: - AnalyticDB-V: - Exqutor:

Q12. “□□ □□□ □□□□□ □□ □□□ □□□□?”

A: Exqutor . ()

:

" $\frac{d}{d}$ $\frac{q}{q}$,

1:

- → (1 가)

2: (Exqutor)

- VGPQ

- -

-

-

Exqutor ECQO .

5. □□ □□ (5 □ □□)

□□□ (30 □)

, . . .

Exqutor(Extended Query Optimizer for Vector-augmented Analytical Queries)
SQL . , AnalyticDB-V .

AnalyticDB-V 2020 VLDB Alibaba Group , VAQ(Vector-Augmented Analytical Query) . , 가 .

□□ □□ □□ (2 □ 20 □)

1. (20)

가 . , , , 가 .

 “ ” , “ 10 ” .

 . AnalyticDB-V SQL .

SELECT * FROM products
WHERE ANNS(image_embedding, query_image) **AND** price < 100000

VAQ .

2. (30)

AnalyticDB-V Lambda .

, . , .

Lambda :- : → - : → - :

VGPQ .

3. VGPQ (20)

VGPQ IVF-PQ . IVF-PQ 가 , .

VGPQ . , - .

2-3 .

4. (30)

AnalyticDB-V .

. :- 100 → 90% → 10ms - 1000 → 95% → 50ms -
 10000 → 99% → 200ms

가 , .

“ 95% ” , 50ms . “ 99% ” , 200ms .

.

Exqutor □□ □□ (1 □ 10 □)

AnalyticDB-V :

AnalyticDB-V . VAQ , ANNS SQL , . Alibaba .

가 :

() .

, “ ” ? 가 “ < 100000 ” . :

- 가 가
- 가 100
- 가
- ...

.

AnalyticDB-V “ 10% ” 가 .

Exqutor 가 .

Exqutor , .

: - (ANNS) - ECQO(Extended Cardinality Query Optimizer) - VG PQ
- 가 - 가

AnalyticDB-V .

□□ □□□□ □□ (30 □)

AnalyticDB-V “ + SQL ” . “ ” .

. :- - - 가

ECQO AnalyticDB-V .

□□□ (30 □)

:

AnalyticDB-V SQL , .

.

Exqutor . ECQO , AnalyticDB-V .

. .

가□ □□

□□ □□ □□

(Embedding): - (,) - : → CNN → 512 - 가ㄴ가가

ANNS (Approximate Nearest Neighbor Search): - , 가 - IVF-PQ, VG PQ,
HNSW -

ANN (Approximate Nearest Neighbor): - ANNS

(Selectivity): - - SELECT COUNT(*) FROM products WHERE price <
1000 / SELECT COUNT(*) FROM products - 0 1 (0 = , 1 =)

(Cardinality): - . ×

□□□□

- AnalyticDB-V : “AnalyticDB-V; A Hybrid Analytical Engine Towards Query Fusion for Structured and Unstructured Data” (VLDB 2020)
- IVF-PQ : “Product Quantization for Nearest Neighbor Search”

- : “A Survey on Vector Database Systems”
-

: 1900 (Markdown) : , - ,